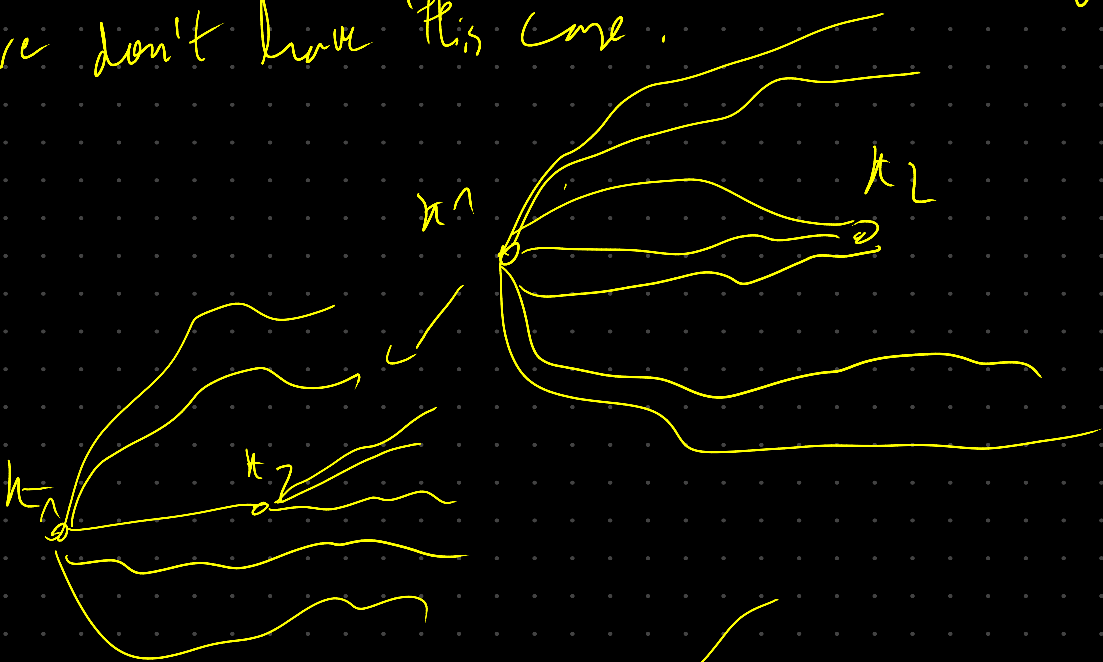
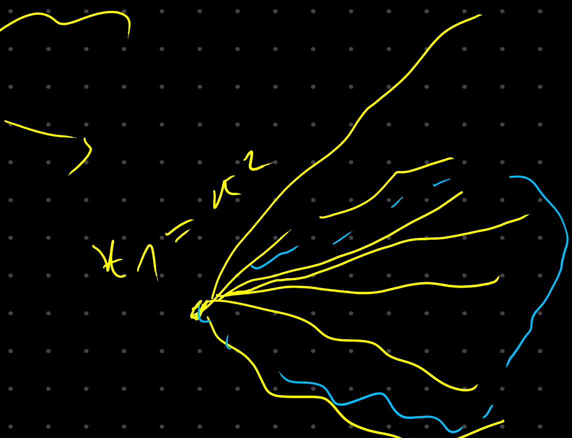
weak H-T: only time, the embedded drawing would be not equivalent to our drawing is when

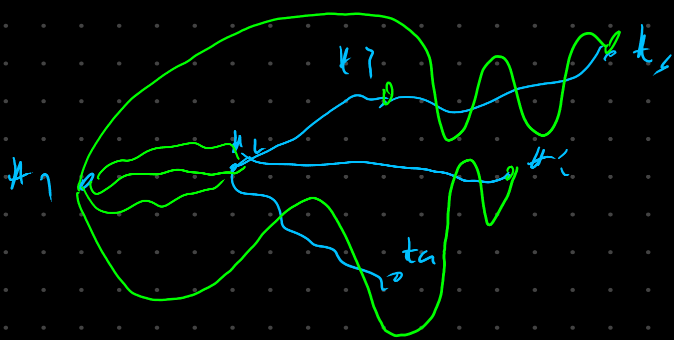
edge case

from this, we know that a weak hanani-tutte drawing leads to an equivalent x-monotone drawing if we don't have this case.



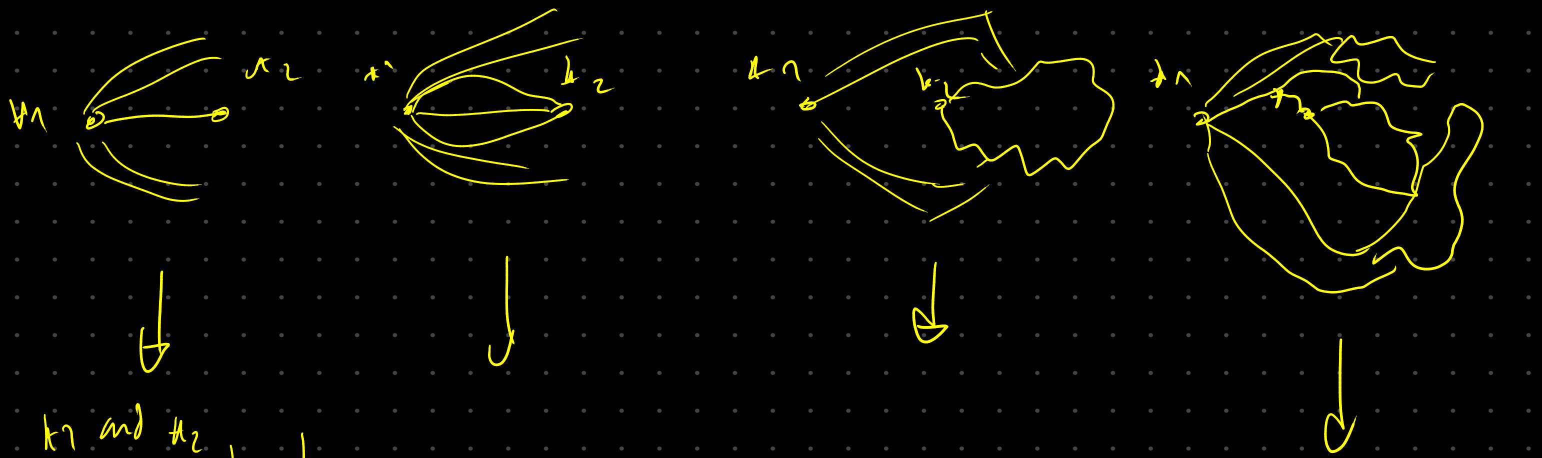


already emb.!

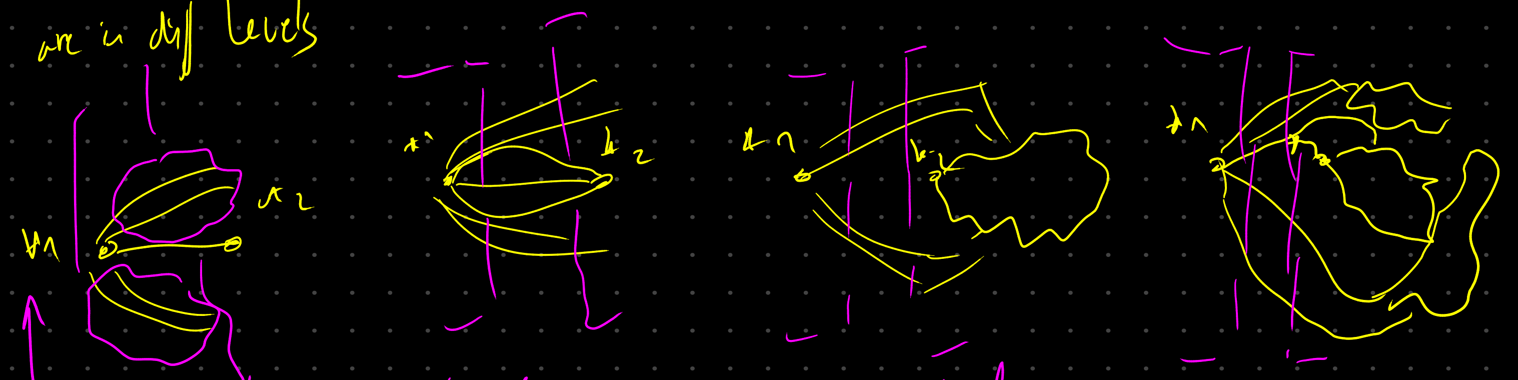


but what

are the case we still have with the H-T drawing from 2SAT,



x1 and x2 are in diff levels



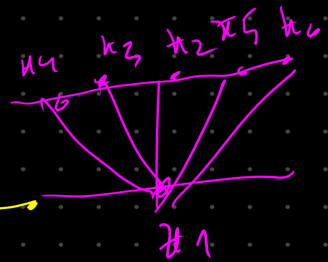
outgoing edge ⇒ can't be a stretched edge

same argument.

x2 can't have a stretch edge to the left of x1 without breaking even crossing

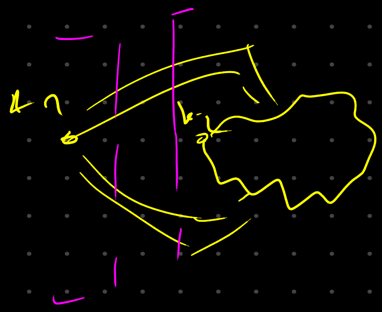
same arg

* shape of subgraphs

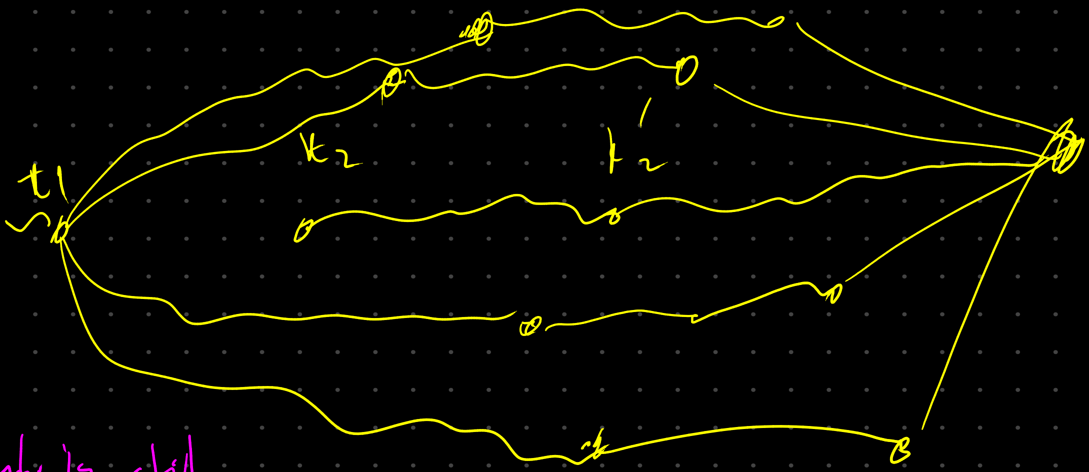


we will come back to this case when we work with a strong H-T.

Same is happening with second case

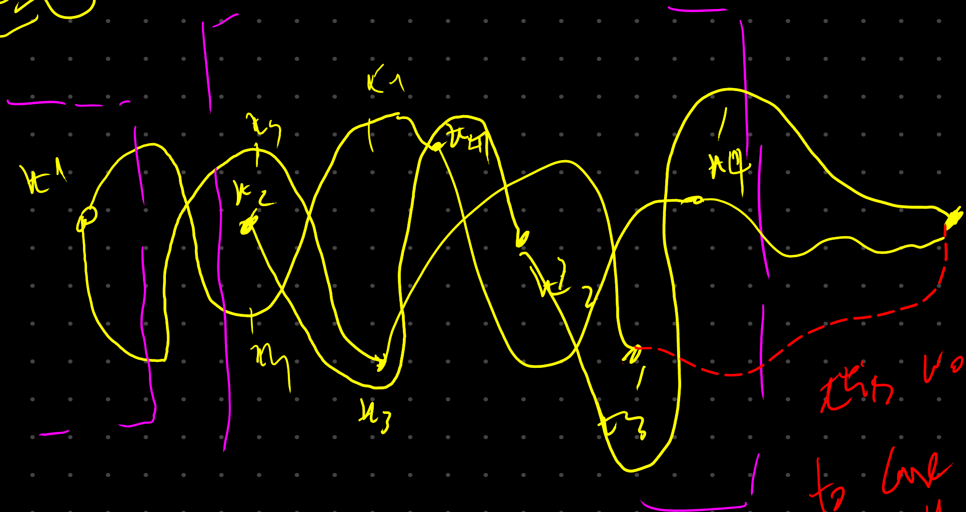


minimal example



emb. is still equivalent to the drawing

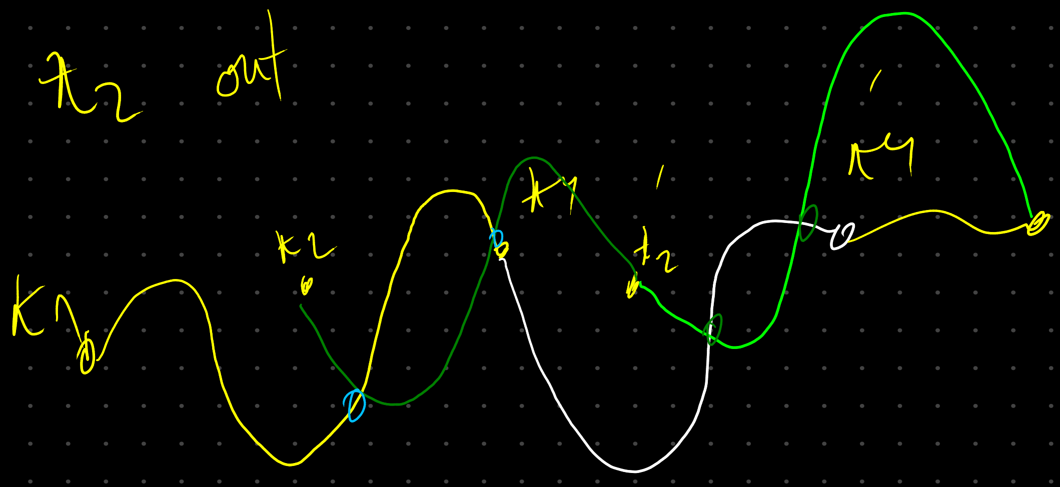
if we take

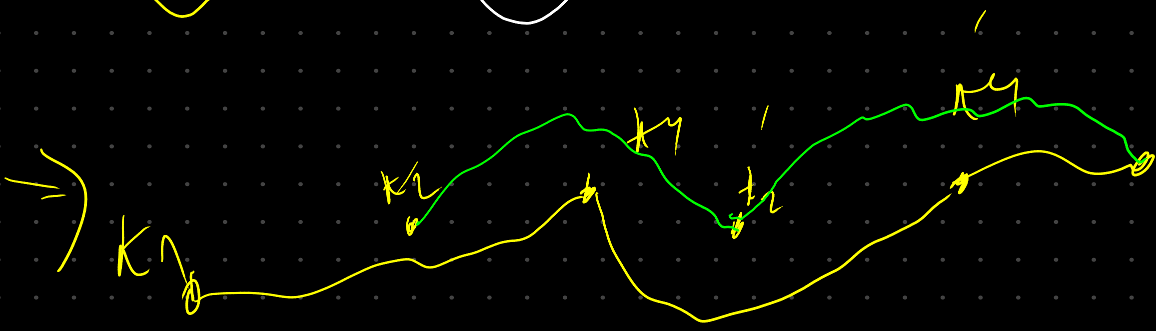


this would lead to one in proof of th 3.1 that tolerates all crossings in adjacent edges.

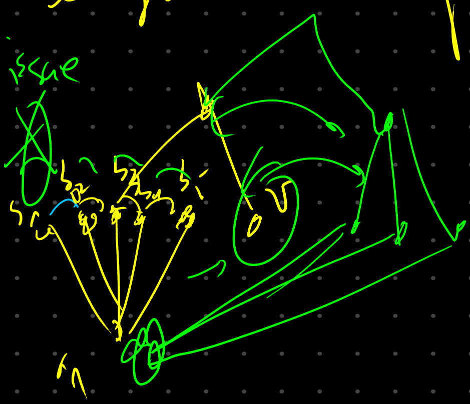
in this case we can change the order so we have

t_2 out



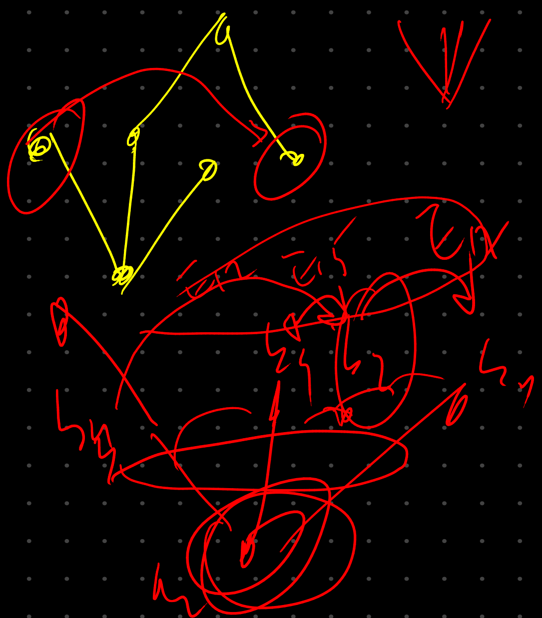


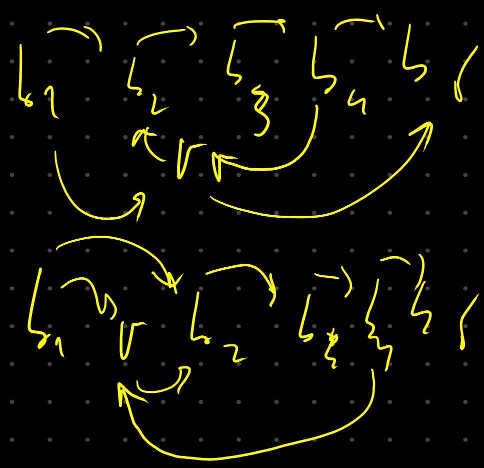
larger example of how we can get into this situation

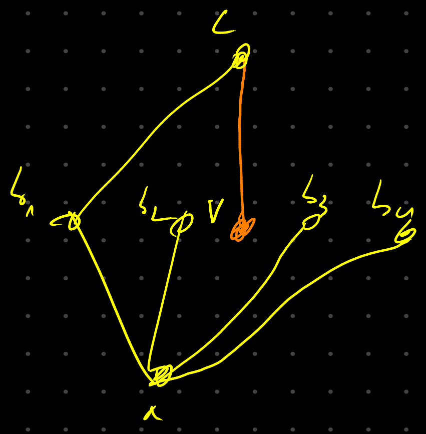


if we are in weak H-T

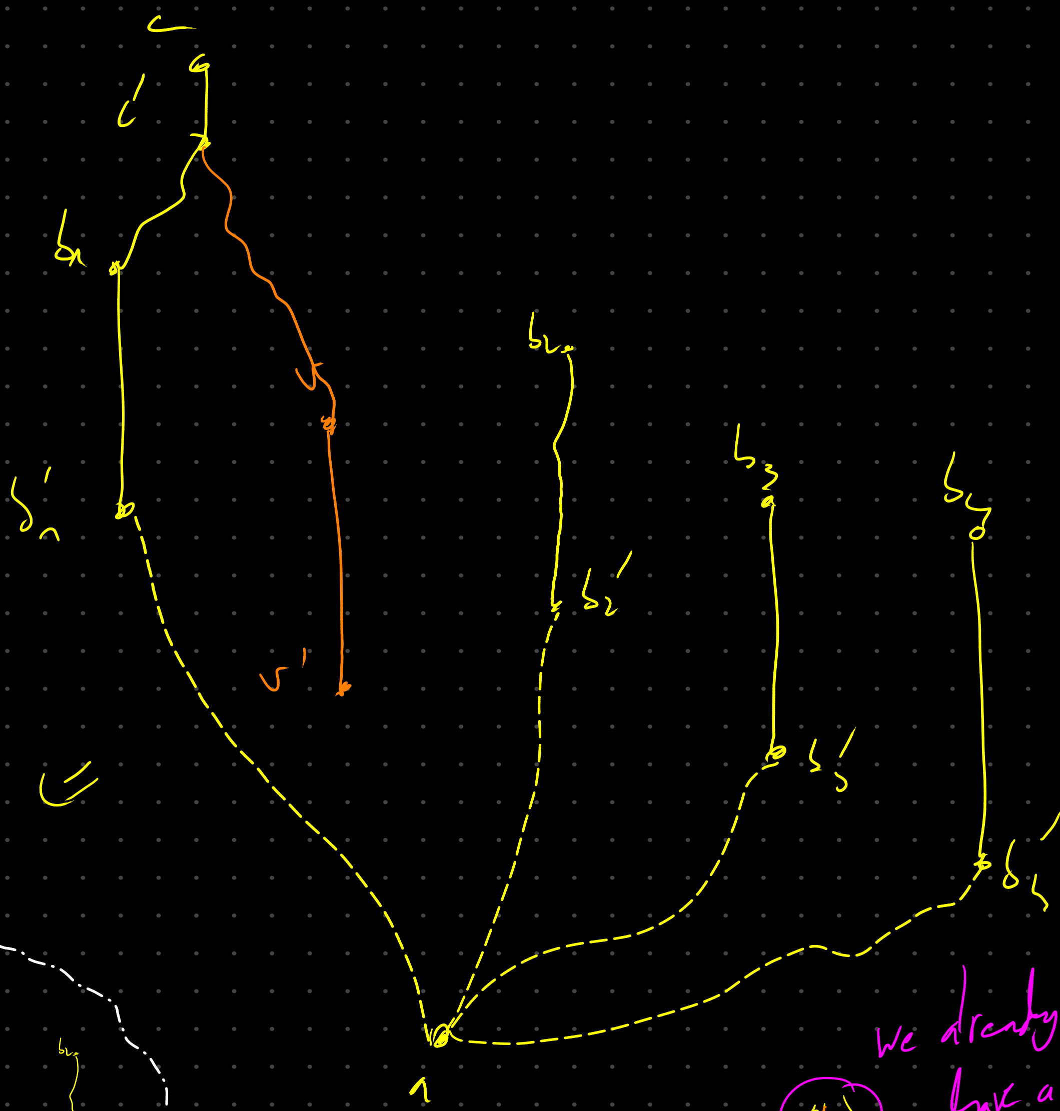
the only issue would be the order of v in regard to others.

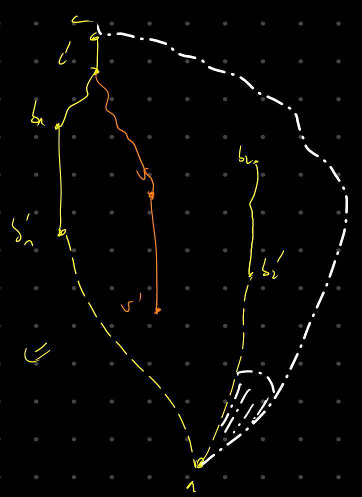




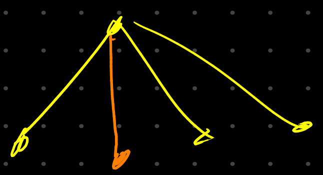


drawing

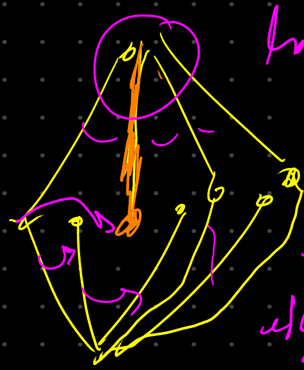




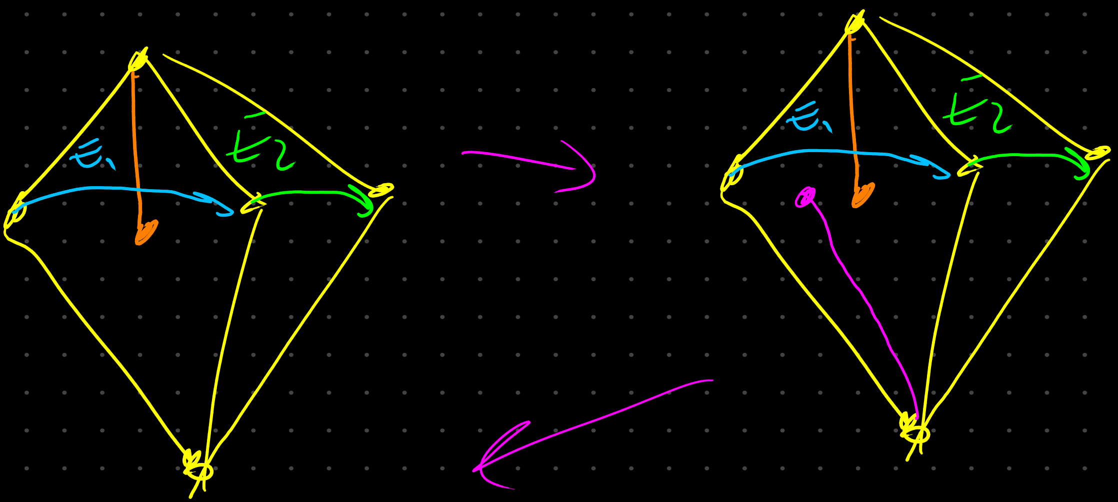
basically redraw

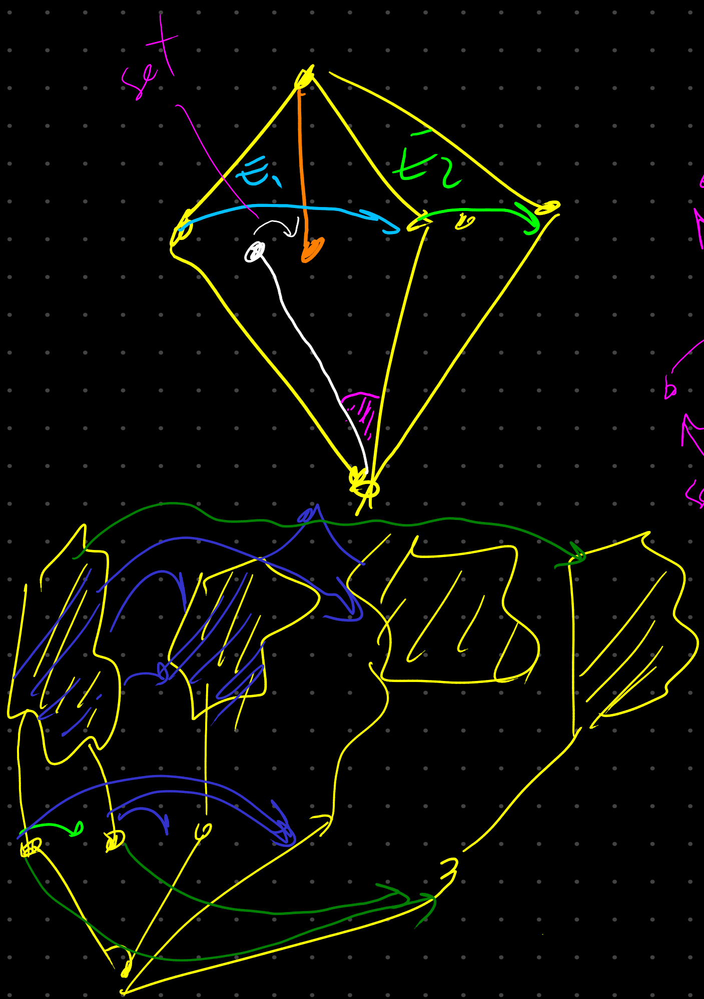


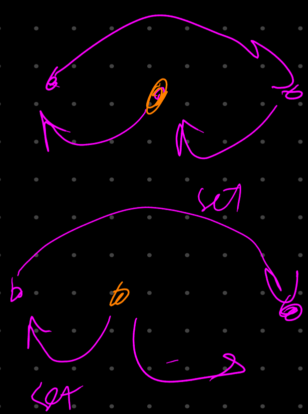
then



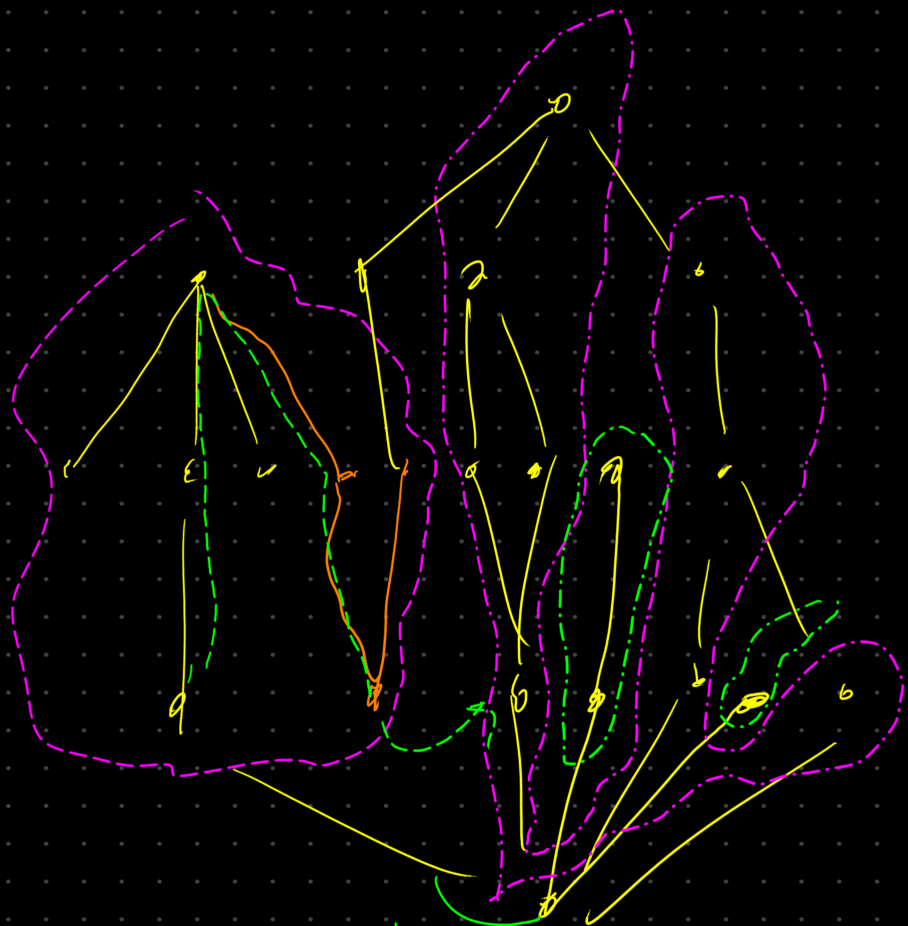
we already have a fixed order between the adjacent edges to our orang edge , we draw the other vertices one by one, taking in consideration their rotations.







we can give same value to all vertices that don't have an edge and linked with our orange vertex.



E_7